

The GLAMSQUAD SALON

Data Cleaning & Preparation Methodology

Profitability & Operational Analytics | Portfolio Methodology

Document Control	Details
Client	The GlamSquad Salon
Role	Business Analyst + Data Analyst
Project Period	May–Aug 2025
Historical Data	Approximately one year
Confirmed Sources	Excel and handwritten operational records
Tools Used	Excel, Google Sheets, SQL, Python and Power BI
Purpose	Document a defensible preparation methodology for converting fragmented salon records into analysis-ready data.
Status	Portfolio reconstruction based on confirmed project facts.
Evidence boundary: The source data is confirmed to have been unclean, but the exact historical issue counts and exact transformation code are not retained. This document therefore separates confirmed project facts from a proposed reproducible portfolio-cleaning methodology.	

1. Purpose & Context

The GlamSquad Salon analysis depended on operational information distributed across Excel files and handwritten records. Before profitability, appointment, customer, service, staff and feedback analysis could be trusted, the available information had to be consolidated, standardized and validated.

The preparation objective was not simply to 'clean a spreadsheet'. It was to create a reliable analytical foundation in which the same customer, service, stylist, appointment status and commercial value could be interpreted consistently across calculations and Power BI reporting.

2. Confirmed Source Domains

Domain	Confirmed Data
Appointments	Appointments/bookings, cancellations and no-shows
Services	Services performed and service prices
Customers	Customer data and repeat visits
Staff	Staff/stylist information, working hours and attendance
Compensation	Salaries and commissions
Costs	Product/material costs and expenses
Commercial	Revenue and discounts
Feedback	Reviews/comments/feedback from multiple channels

3. Preparation Workflow

End-to-end flow: Source inventory → digitize/consolidate records → profile data → standardize structure → resolve duplicates/mappings → validate statuses and commercial fields → create analytical keys/derived fields → reconcile totals → prepare model-ready tables → validate Power BI outputs.

4. Step 1 — Source Inventory & Intake

The first step is to understand what information exists, where it is stored and which business question each source can support.

Check	Methodology
File/source inventory	Identify available Excel files, sheets and handwritten record categories.
Period coverage	Confirm which dates are represented and whether sources overlap.
Business ownership	Confirm with the manager what each source represents operationally.
Grain	Determine whether a row represents an appointment, service, customer, staff record, expense or another business event.
Field meaning	Clarify ambiguous headings or handwritten labels before transformation.
Confidentiality	Identify customer/staff/commercial fields that must not appear in public portfolio outputs.
Do not assume: If the meaning of a source field cannot be established, document it as unresolved or exclude it from the affected calculation rather than inventing a definition.	

5. Step 2 — Handwritten-to-Digital Consolidation

Handwritten operational information must be transcribed into a controlled tabular structure before it can be joined with Excel data. The historical transcription volume is not retained, so no record count is claimed.

- Use consistent column headings for the target table.
- Enter dates, service names, staff names, statuses and numeric values into separate fields rather than combined free text.
- Preserve the original source reference or a portfolio source tag where useful for reconciliation.
- Perform a second-pass review of manually transcribed values, particularly dates and commercial amounts.
- Do not infer unreadable or missing values; mark them for review.

6. Step 3 — Data Profiling

Before applying transformations, profile each dataset to identify data-quality risks.

Profiling Area	Checks
Completeness	Null/blank values in key analytical fields.
Uniqueness	Potential duplicate appointment/customer/expense records.
Validity	Invalid dates, non-numeric amounts, impossible/unsupported statuses.
Consistency	Variations in service names, stylist names and categorical values.
Range / reasonableness	Values requiring business review because they appear inconsistent with surrounding records.
Referential integrity	Appointments referencing customers/services/staff that cannot be mapped.
Time coverage	Missing periods or unexpected gaps that could distort trends.

7. Step 4 — Structural Standardization

The cleaned analytical structure should separate dimensions and facts so that repeated business labels are standardized before aggregation.

Field Type	Preparation Rule
Dates	Convert to a consistent date format and reject/flag unparseable values.
Times	Standardize time representation where available for demand analysis.
Service names	Trim whitespace, normalize case/spelling and map equivalent labels to one standard service.
Stylist/staff names	Normalize spelling/format and map variants to a single sanitized staff key.
Appointment statuses	Map source variants into controlled analytical categories such as completed/attended, cancelled and no-show only when supported.
Numeric fields	Convert revenue, discounts, expenses, salary/commission and material costs to valid numeric types.
Text fields	Trim unnecessary spaces and normalize blanks/nulls without changing business meaning.

8. Step 5 — Customer Identity & Repeat-Visit Preparation

Confirmed logic: Phone-number data was used to identify repeat customers/visits.

For the portfolio rebuild, customer matching should preserve that confirmed logic while protecting PII.

- Normalize phone-number formatting before matching where the source requires it.
- Use the normalized phone value only in the protected preparation layer.
- Create a sanitized Customer_Key or masked/hashed matching key for analytical use.
- Check for duplicate customer records associated with the same matching value.
- Count validated visits per logical customer to support repeat-customer analysis.
- Remove or mask phone numbers from all public portfolio datasets and visuals.

Caution: A shared or incorrectly entered phone number can create false matches. Ambiguous matches should be reviewed or documented rather than silently merged.

9. Step 6 — Appointment Status Preparation

Cancellation and no-show analysis depends on consistent appointment outcomes.

Preparation	Control
Status normalization	Map spelling/case variants to controlled status values.
Completed/attended logic	Include only source statuses that clearly represent delivered appointments.
Cancelled logic	Use source evidence that the booking was cancelled.
No-show logic	Use source evidence that the customer did not attend.
Unknown status	Keep unresolved statuses separate until clarified.
Rate denominator	Document the eligible/booked denominator before calculating cancellation/no-show percentages.

10. Step 7 — Service, Pricing & Discount Preparation

- Standardize service names before grouping revenue or appointment counts.
- Validate service prices and transaction revenue fields as numeric values.
- Keep listed price, actual revenue and discount fields conceptually separate where the rebuild data supports them.
- Check whether discount values are amounts or percentages before calculation.
- Do not infer a discount when the source does not provide sufficient evidence.
- Prepare service-level direct costs only where product/material or commission attribution is defensible.

Commercial confidentiality: Original service prices, discount percentages, margins and revenue values are not published in the portfolio.

11. Step 8 — Staff, Working-Hours & Compensation Preparation

Area	Methodology
Staff identity	Standardize staff/stylist labels and assign sanitized analytical keys.
Working hours	Convert available working-hour information into a consistent numeric/time basis.
Attendance	Standardize attendance categories where used.
Appointments	Map validated appointments/services to the responsible stylist where available.
Revenue	Attribute revenue to stylists only where the source relationship is supported.
Salary / commission	Validate numeric values and reporting period before comparing them with performance.
Utilization	Calculate only if a defensible working-capacity and used/booked-time denominator exists.

12. Step 9 — Revenue, Expense & Cost Preparation

Profitability analysis requires consistent treatment of revenue and expenses.

- Validate revenue and expense values as numeric fields.
- Standardize expense categories where required for analysis.
- Confirm the reporting date/period associated with each commercial record.
- Separate direct service costs from general expenses only where the source supports that distinction.
- Avoid double counting expenses represented in more than one source.
- Reconcile aggregated revenue and expense totals before calculating profit.
- Use Profit = Revenue – Expenses at the agreed business reporting level.

13. Step 10 — Feedback Preparation

Qualitative feedback came from Google reviews, Instagram comments, WhatsApp feedback, surveys, salon records and direct conversations.

Preparation Step	Methodology
Collect / reference	Maintain source/channel context where available.
Sanitize	Remove customer-identifying information from portfolio material.
Review	Read feedback for recurring operational/business themes.
Categorize	Assign consistent themes where evidence supports classification.
Confirmed themes	Pricing, staff behaviour, cleanliness and staff availability.
Separate evidence types	Do not combine feedback counts with transactional revenue/appointment totals as if they were the same data type.
No invented sentiment: Do not claim historical sentiment scores, positive/negative percentages or review counts unless they are actually calculated from a retained/sanitized dataset.	

14. Step 11 — Derived Analytical Fields

Derived Field	Portfolio Preparation Logic	Evidence Status
Customer_Key	Sanitized identifier representing a logical customer.	Proposed rebuild field
Repeat_Customer_Flag	Derived from validated visit count under the phone-based matching logic.	Rebuild implementation of confirmed analysis
Appointment_Status_Standard	Controlled mapping from validated source statuses.	Proposed standardized field
Profit	Revenue – Expenses.	Confirmed high-level logic
Service_Contribution	Service Revenue – reliably attributable Direct Service Costs.	Conditional rebuild logic
Revenue_per_Working_Hour	Attributed stylist revenue ÷ validated working hours.	Portfolio KPI logic
Staff_Utilization_Pct	Used/booked service time ÷ available working time, only where denominator exists.	Conditional portfolio logic
Feedback_Theme	Category assigned to sanitized qualitative feedback.	Rebuild implementation of confirmed feedback analysis

15. Step 12 — Model-Ready Dataset Preparation

After cleaning, the data can be organized into a logical portfolio model such as Customers, Appointments, Services, Staff, Staff Working Hours, Expenses, Feedback and Date tables.

- Use unique sanitized keys for dimensions.
- Keep appointment/service activity at a consistent transaction grain.
- Use a Date dimension for time-based reporting.
- Validate one-to-many mappings before loading to Power BI.
- Do not create relationships solely to make a visual work if the business relationship is not defensible.

16. Validation & Reconciliation Framework

Validation	Expected Check
Record-count check	Compare source and prepared counts by relevant source/period while accounting for documented exclusions.
Appointment reconciliation	Check completed/cancelled/no-show classifications against cleaned source statuses.
Customer check	Review duplicate/matching logic before repeat-customer calculations.
Service mapping	Confirm all reportable services map to the intended standardized service.
Staff mapping	Confirm stylist/staff labels map consistently across appointments and working-hour data.
Revenue reconciliation	Compare prepared revenue totals with the cleaned source/control totals.
Expense reconciliation	Compare prepared expense totals with the cleaned source/control totals.
Profit check	Confirm Profit = Revenue – Expenses at the same filter/reporting context.
Power BI check	Reconcile major dashboard KPIs to prepared analysis outputs.
Privacy check	Verify public files contain no phone numbers or confidential client values.

17. Missing / Ambiguous Data Handling

Situation	Treatment
Missing key value	Flag and assess whether the record can support the intended analysis.
Unknown appointment status	Do not force into completed/cancelled/no-show.
Unclear handwritten value	Mark for review; do not guess.
Duplicate candidate	Investigate using available business fields before removal/merge.
Missing direct cost	Exclude from service-margin calculation or clearly state limitation.
Missing capacity denominator	Do not publish a utilization percentage; use supported activity views instead.
Ambiguous customer match	Do not silently merge customers.
Outlier	Validate against source/business context before correcting or excluding.

18. Tool Usage

Tool	Role in Methodology
Excel	Source inspection, reconciliation, spot checks and structured cleaning tasks.
Google Sheets	Working with tabular business information and collaborative/operational handling where applicable.
Python	Reproducible cleaning, profiling, transformations and exploratory analysis.
SQL	Structured validation, filtering, aggregation and analytical checks.
Power BI	Data model, KPI measures, visual analysis and management reporting.
Historical code boundary: The exact original Python scripts, SQL queries and Power BI transformation steps are not retained. Future code in the portfolio should be labeled as a reproducible rebuild, not historical client code.	

19. Data-Quality Checklist

- Source inventory completed and source purpose understood.
- Handwritten records transcribed/reviewed without guessing unreadable values.
- Dates and times standardized.
- Service names standardized.
- Staff/stylist labels standardized.
- Appointment statuses mapped and unresolved values retained separately.
- Customer repeat matching reviewed and phone numbers removed from public output.
- Revenue, discounts, expenses, salary/commission and material-cost fields validated.
- Duplicate candidates reviewed.
- Direct-cost attribution used only where defensible.
- Revenue and expense totals reconciled.
- Profit calculation checked.
- Power BI KPIs reconciled with prepared data.
- Customer/staff/commercial confidentiality reviewed before publication.

20. Confirmed vs Portfolio-Reconstructed Methodology

Classification	What Is Supported
Confirmed	Approximately one year of unclean data; Excel and handwritten sources; listed business data domains; phone number used for repeat visits; tools included Excel, Google Sheets, SQL, Python and Power BI; dashboard built from scratch.
Confirmed analysis need	Profitability, appointments/utilization, repeat customers/retention, service mix/pricing, staff performance/scheduling, costs, cancellations/no-shows and customer feedback.
Portfolio reconstruction	Exact cleaning sequence, logical field names, standardized status values, sanitized keys, validation checklist and model-ready table design documented here.
Not retained / not claimed	Exact historical issue counts, duplicate counts, null percentages, original table/column names, exact Python/SQL cleaning code, exact Power Query steps or exact historical transformation thresholds.